

Xiangling Xu (许湘灵)

My Website

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G Google Scholar

ID Orcid



Research Interests: Quantum information, quantum foundations, operator algebras, noncommutative polynomials optimization

Education

- 2023 – now **Ph.D., Inria Saclay Île-de-France & École doctorale IP Paris**
Supervisor: Marc-Olivier Renou
Project: *Quantum correlations in causal structures: characterization and application*
- 2020 – 2023 **M.Sc. Mathematics, ETH Zürich**
Master Thesis: *Quantum Nonlocality in Bilocal Networks: An Operator Algebraic Perspective*
Semester Project: *Quantum BCOV Theory on Calabi-Yau Manifold*
CGPA: 5.69/6.00
- 2016 – 2020 **B.Sc. Mathematics and Physics, University of Toronto**
Projects: *Noncommutative Geometry in Gravity, Large-Scale Cosmological Structures, Optical Simulation, Biomechanical Simulation*
CGPA: 3.95/4.00

Publications and Preprints

- 1 M. Baroni, I. Klep, D. Leichtle, M.-O. Renou, I. Šupić, L. Tendick, and **X. Xu**, *Quantitative quantum soundness for all multipartite compiled nonlocal games*, 2025. arXiv: 2509.25145 [quant-ph]. [URL: https://arxiv.org/abs/2509.25145](https://arxiv.org/abs/2509.25145).
- 2 I. Klep, C. Paddock, M.-O. Renou, S. Schmidt, L. Tendick, **X. Xu**, and Y. Zhao, *Quantitative quantum soundness for bipartite compiled bell games via the sequential npa hierarchy*, 2025. arXiv: 2507.17006 [quant-ph]. [URL: https://arxiv.org/abs/2507.17006](https://arxiv.org/abs/2507.17006).
- 3 **X. Xu**, M.-O. Renou, and I. Klep, *Quantitative tsirelson's theorems via approximate schur's lemma and probabilistic stampfli's theorems*, 2025. arXiv: 2505.22309 [quant-ph]. [URL: https://arxiv.org/abs/2505.22309](https://arxiv.org/abs/2505.22309).
- 4 M.-O. Renou, **X. Xu**, and L. T. Ligthart, *Two convergent npa-like hierarchies for the quantum bilocal scenario*, 2024. arXiv: 2210.09065 [quant-ph]. [URL: https://arxiv.org/abs/2210.09065](https://arxiv.org/abs/2210.09065).
- 5 X. Luo, **X. Xu**, and X. Wang, "On the kinematic morphology around haloes," *Monthly Notices of the Royal Astronomical Society*, vol. 518, no. 4, pp. 6059–6064, 2023. [DOI: https://doi.org/10.1093/mnras/stac3500](https://doi.org/10.1093/mnras/stac3500).
- 6 **X. Xu**, "Quantum nonlocality in bilocal networks: An operator algebraic perspective," M.S. thesis, ETH Zurich, 2023. [DOI: https://doi.org/10.3929/ethz-b-000613670](https://doi.org/10.3929/ethz-b-000613670).

Talks & Conference Activities

Invited Talks

- Jun 2026 **2026 SIAM Conference on Optimization (OP26)**, Edinburgh, United Kingdom

Jul 2025  *IWOTA 2025 Twente*, Twente, Netherland

Contributed Talks

Jan 2026	📖	<i>QIP 2026 Riga</i> – 2 contributed talks, Riga, Latvia
Nov 2025	📖	<i>3rd Colloquium GDR TeQ</i> , Grenoble, France
Oct 2025	📖	<i>YQIS25</i> , Barcelona, Spain
Jun 2025	📖	<i>IQC-PCQT-Quantum Saclay Workshop</i> , Paris, France
Sep 2024	📖	<i>Causalworlds 2024</i> , Waterloo, Canada
Nov 2023	📖	<i>Workshop Defi EQIP</i> , Lyon, France
Oct 2023	📖	<i>Saclay Quantum Seminar</i> , Inria Saclay, France

Poster Sessions

Nov 2024  *2nd Colloquium GDR TeQ*, Paris, France
 YQIS24, Paris, France

Other Participation

Mar 2025	828. WE-Heraeus-Seminar: Operator Theory and Polynomial Optimization in Quantum Information Theory, Bad Honnef, Germany
Sep 2024	VQF Vienna Quantum Foundations Conference, Vienna, Austria
Nov 2023	POP23 Workshop on Future Trends in Polynomial OPTimization, Toulouse, France

Collaborations

Invited Group Visits

Oct-Nov 2025		<i>Thomas Vidick</i> , EPFL, Lausanne Invited two-week visit
Sep 2024		<i>Igor Klep</i> , University of Ljubljana Invited two-week visit funded by Erasmus+ Mobility program
Jul 2024		<i>William Slofstra</i> , Institute for Quantum Computing, University of Waterloo Invited two-week visit
Dec 2023		<i>David Gross</i> , University of Cologne Invited one-week visit

Organizations

Mar 2025 - Now	- Coorganizer for <i>Seminars in Commutative and Noncommutative Polynomial Optimization for Quantum Theory</i> - Coordinating the Gaspard Monge Visiting Professor program for Igor Klep - Co-planning the seminar and the schedule for invited speakers
Mar 2025	- Coorganizer for <i>828. WE-Heraeus-Seminar at Bad Honnef</i> - Exploring Operator Theory and Polynomial Optimization in Quantum Information Theory with world-renowned experts - Coordinating with the organizers and inviting some of the speakers
Feb 2024 - Now	- Coorganizer of <i>Project COMPUTE for Quanterra</i> - Coordinating with the members - Making the project poster and website

Collaborations (continued)

- Sep 2023  Coorganizer for *Saclay Quantum Seminar on NPA hierarchy*
Planning and organizing the one-day program to bring all quantum groups at Saclay together

Awards and Scholarships

- 2024  *Erasmus+ Mobility Program*, funds for visiting the University of Ljubljana
- 2020  *McNab Undergraduate Scholarship*, University of Toronto
- 2017 - 2020  *Dean's List Scholar*, University of Toronto
- 2018  *The Class of 3To and Associates Scholarship in Mathematics and Physics*, University of Toronto
-  *International Exchange Award*, University of Toronto
-  *Bronze Medal*, 2017 University Physics Competition
- 2017  *Eileen Doman Award*, University of Toronto
-  *Summer Undergraduate Research Program*, Canadian Institute for Theoretical Astrophysics
- 2016  *President's Scholar of Excellence Program*, University of Toronto

Teaching Experience

Lecture

- Nov 2024  *Recent trends in parallel, distributed, and quantum computing*
Master 2 QDCS, Université Paris-Saclay

Teaching Assistance

- 2022-2023  *Linear Algebra I & II*
Department of Mathematics, ETH Zürich
- 2021-2022  *Analysis I & II*
Department of Mathematics, ETH Zürich

Skills

- Languages  Chinese (native), English (fluent), German (beginner)
- Coding  Python (Ncpol2sdpa), Matlab (Yalmip), Mathematica, C Language, Hugo on GitHub, $\LaTeX$

Referees

Available on Request